

Date: 30/07/2025

Lab Practical #09:

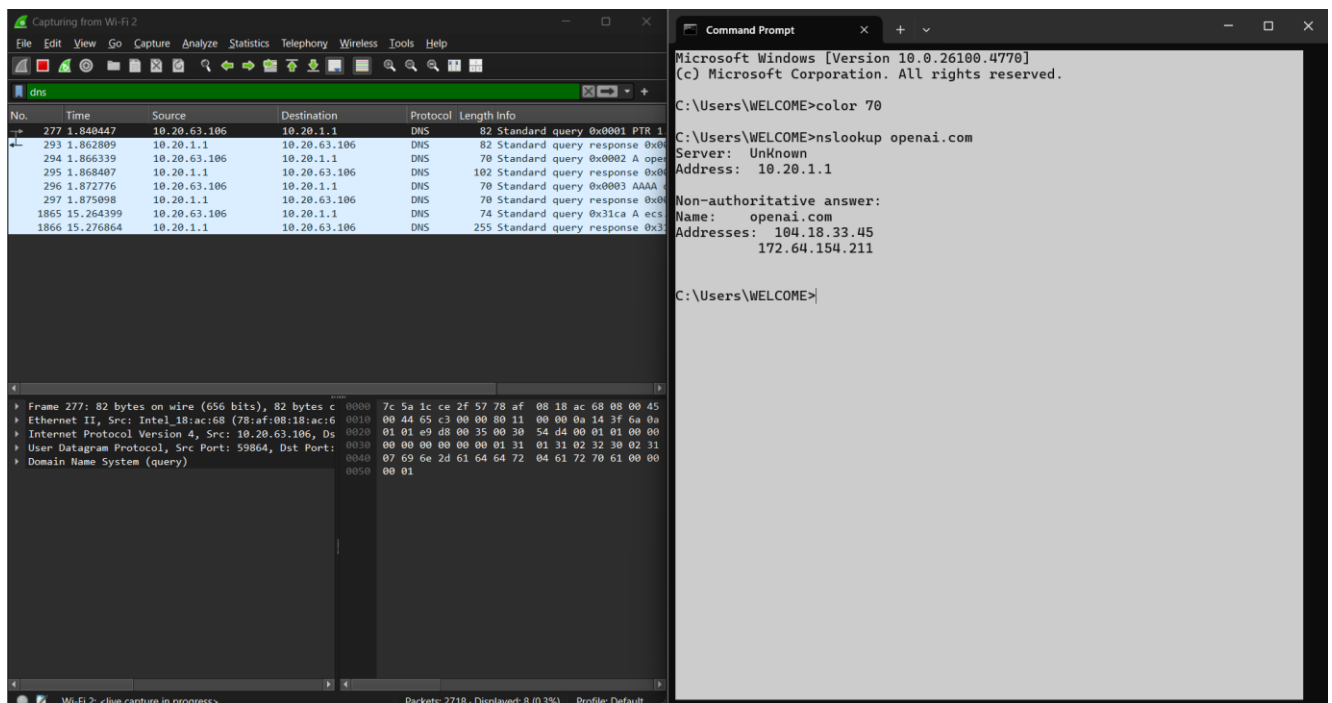
Study Packet capture and header analysis by Wireshark (HTTP, TCP, UDP, IP, etc.)

Practical Assignment #09:

1. Explain usage of Wireshark tool.

Wireshark is a network protocol analyzer used to capture and analyze network traffic in real time. It helps in troubleshooting network issues, monitoring performance, detecting security threats, and learning protocol behavior.

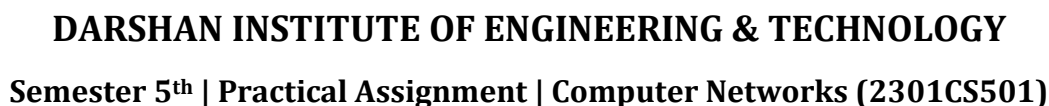
2. Packet capture and header analysis by Wireshark (HTTP, TCP, UDP, IP, etc.)



The screenshot displays two windows side-by-side. The left window is Wireshark, showing a packet capture from 'Wi-Fi 2'. The packet list shows a DNS query (No. 277) and its response (No. 293). The packet details pane shows the structure of the DNS query and response. The right window is a Command Prompt, showing the execution of the command 'nslookup openai.com', which returns the IP addresses 104.18.33.45 and 172.64.154.211.

No.	Time	Source	Destination	Protocol	Length	Info
277	1.840447	10.20.63.106	10.20.1.1	DNS	82	Standard query 0x0001 PTR 1
293	1.862809	10.20.1.1	10.20.63.106	DNS	82	Standard query response 0x0001
294	1.866339	10.20.63.106	10.20.1.1	DNS	70	Standard query 0x0002 A openai.com
295	1.868407	10.20.1.1	10.20.63.106	DNS	102	Standard query response 0x0002
296	1.872776	10.20.63.106	10.20.1.1	DNS	70	Standard query 0x0003 AAAA openai.com
297	1.875098	10.20.1.1	10.20.63.106	DNS	70	Standard query response 0x0003
1865	15.264399	10.20.63.106	10.20.1.1	DNS	74	Standard query 0x31ca A openai.com
1866	15.276864	10.20.1.1	10.20.63.106	DNS	255	Standard query response 0x31ca

```
Microsoft Windows [Version 10.0.26100.4770]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\WELCOME>color 70  
  
C:\Users\WELCOME>nslookup openai.com  
Server: UnKnown  
Address: 10.20.1.1  
  
Non-authoritative answer:  
Name: openai.com  
Addresses: 104.18.33.45  
172.64.154.211  
  
C:\Users\WELCOME>
```



The image is a composite of two screenshots. The top screenshot shows a Wireshark network traffic capture. The packet list pane on the left shows a series of HTTP requests and responses. The packet details pane in the center shows the structure of a selected packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol. The packet bytes pane on the right shows the raw data of the selected packet. The bottom screenshot shows a web browser window displaying a Sophos login page. The page has a dark blue header with the Sophos logo and navigation links. The main content area has a white background with a green checkmark and the text 'You are signed in as 2301010299'. Below this, there is a green button labeled 'Sign out' and a link labeled 'Access the User Portal'.



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Semester 5th | Practical Assignment | Computer Networks (2301CS501)

Date: 30/07/2025

Wireshark packet capture showing ICMP Echo (ping) requests and replies. The packet list shows 8 packets, and the packet details pane shows the structure of an ICMP Echo request and reply.

Command Prompt window showing the execution of 'color 70' and 'ping google.com'. The ping command shows statistics for google.com [142.250.207.142] with 32 bytes of data, 4 packets sent, 4 received, 0% loss, and round trip times ranging from 32ms to 48ms.

Wireshark packet capture showing TCP connections between 10.20.63.106 and 10.20.63.106. The packet list shows 24 packets, and the packet details pane shows the structure of a TCP connection establishment (SYN, SYN-ACK, ACK) and data transfer.

Screenshot of the Darshan University Learning Management System (LMS) dashboard. The dashboard shows the user's profile (Kalp Vipul Viradia) and a list of courses for Semester 5, including 2301CS402 - Design and Analysis of Algorithm and 2301CS501 - Computer Network.



DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

Semester 5th | Practical Assignment | Computer Networks (2301CS501)

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The image displays a Wireshark network traffic analysis window. The top pane shows a list of captured packets, with the selected packet (No. 246) being a TCP Reset (RST) from 10.20.63.106 to 10.255.1.1. The middle pane shows the details of the selected packet, including the Ethernet II header, Internet Protocol Version 4 header, and Transmission Control Protocol header. The bottom pane shows the raw packet data in hexadecimal and ASCII. In the background, a YouTube video player is visible, showing a video titled "I Tried India's First Tesla! *Tesla Model Y*" by Tech Burner.

No.	Time	Source	Destination	Protocol	Length	Info
16	0.128272	10.20.63.106	148.113.16.94	TCP	55	50569 → 443 [ACK] Seq=1 Ack=1 Win=254 Len=1
17	0.151965	148.113.16.94	10.20.63.106	TCP	66	443 → 50569 [ACK] Seq=1 Ack=2 Win=501 Len=0 SLE=1 SRE=2
230	1.668014	10.20.63.106	8.8.4.4	TLSv1.2	243	Application Data
231	1.668068	10.20.63.106	8.8.4.4	TLSv1.2	93	Application Data
232	1.668097	10.20.63.106	8.8.4.4	TLSv1.2	244	Application Data
233	1.680166	8.8.4.4	10.20.63.106	TCP	60	443 → 50773 [ACK] Seq=1 Ack=190 Min=1044 Len=0
234	1.680166	8.8.4.4	10.20.63.106	TCP	60	443 → 50773 [ACK] Seq=1 Ack=229 Min=1044 Len=0
235	1.680166	8.8.4.4	10.20.63.106	TCP	60	443 → 50773 [ACK] Seq=1 Ack=419 Min=1044 Len=0
236	1.680166	8.8.4.4	10.20.63.106	TLSv1.2	93	Application Data
237	1.681143	8.8.4.4	10.20.63.106	TLSv1.2	161	Application Data
238	1.681143	8.8.4.4	10.20.63.106	TLSv1.2	553	Application Data
239	1.681143	8.8.4.4	10.20.63.106	TLSv1.2	85	Application Data
240	1.681143	8.8.4.4	10.20.63.106	TLSv1.2	93	Application Data
241	1.681143	8.8.4.4	10.20.63.106	TLSv1.2	139	Application Data
242	1.681143	8.8.4.4	10.20.63.106	TLSv1.2	553	Application Data
243	1.681143	8.8.4.4	10.20.63.106	TLSv1.2	85	Application Data
244	1.681214	10.20.63.106	8.8.4.4	TCP	54	50773 → 443 [ACK] Seq=419 Ack=1331 Win=250 Len=0
245	1.682016	10.20.63.106	8.8.4.4	TLSv1.2	89	Application Data
246	1.682091	10.20.63.106	8.8.4.4	TLSv1.2	93	Application Data
247	1.682597	10.20.63.106	10.255.1.1	TCP	54	50775 → 8090 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	fe80::19dc:7cce:898...	ff02::1:3	LLMNR	85	Standard query 0x9f56 A 16
3	0.000340	10.20.19.10	224.0.0.252	LLMNR	65	Standard query 0x9f56 A 16
4	0.000537	fe80::19dc:7cce:898...	ff02::1:3	LLMNR	85	Standard query 0xb7ca AAA
5	0.000632	10.20.19.10	224.0.0.252	LLMNR	65	Standard query 0xb7ca AAA
6	0.000916	10.20.62.58	224.0.0.251	MDNS	79	Standard query 0x972c PTR
7	0.102534	fe80::1cbf:d5bb:d82...	ff02::1:3	LLMNR	88	Standard query 0xd51b A 05
8	0.102748	10.20.27.8	224.0.0.252	LLMNR	68	Standard query 0xd51b A 05
9	0.102844	fe80::1cbf:d5bb:d82...	ff02::1:3	LLMNR	88	Standard query 0xa157 AAA
11	0.204849	10.20.63.211	224.0.0.251	MDNS	166	Standard query 0x0000 PTR
12	0.205062	fe80::18b3:39b:fdfe...	ff02::fb	MDNS	186	Standard query 0x0000 PTR
13	0.205385	10.20.62.58	224.0.0.251	MDNS	102	Standard query 0x6d0e PTR
17	0.307888	fe80::d405:5f99:3c6...	ff02::1:3	LLMNR	85	Standard query 0xfa8e A 16
18	0.307888	10.20.19.16	224.0.0.252	LLMNR	65	Standard query 0xfa8e A 16
19	0.307888	fe80::d405:5f99:3c6...	ff02::1:3	LLMNR	85	Standard query 0x4bf0 AAA
20	0.308197	10.20.19.16	224.0.0.252	LLMNR	65	Standard query 0x4bf0 AAA
21	0.308249	10.20.63.183	224.0.0.251	MDNS	117	Standard query response 0x
22	0.308607	fe80::845:3c59:67cc...	ff02::fb	MDNS	137	Standard query response 0x
30	0.409759	fe80::ee6a:1e86:d28...	ff02::1:3	LLMNR	85	Standard query 0x1d7b A 10
31	0.409918	10.20.19.7	224.0.0.252	LLMNR	65	Standard query 0x1d7b A 10



The image displays a Wireshark packet capture of network traffic. The top bar indicates the capture is on the 'Wi-Fi 2' interface. The packet list on the left shows a series of DNS queries and responses. The packet details pane on the right shows the structure of a DNS query, including the question section with a PTR record for 'PTR_arduino_tcp.local'. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	fe80::19dc:7cce:898...	ff02::1:3	LLMNR	85	Standard query 0x9f56 A 10_04
3	0.000340	10.20.19.10	224.0.0.252	LLMNR	65	Standard query 0x9f56 A 10_04
4	0.000537	fe80::19dc:7cce:898...	ff02::1:3	LLMNR	85	Standard query 0xb7ca AAAA 10_04
5	0.000632	10.20.19.10	224.0.0.252	LLMNR	65	Standard query 0xb7ca AAAA 10_04
6	0.000916	10.20.62.58	224.0.0.251	MDNS	79	Standard query 0x972c PTR_arduino_tcp.local, "QM" question
7	0.102534	fe80::1cbfd5bb:d82...	ff02::1:3	LLMNR	88	Standard query 0xd51b A 05_17_08
8	0.102748	10.20.27.8	224.0.0.252	LLMNR	68	Standard query 0xd51b A 05_17_08
9	0.102844	fe80::1cbfd5bb:d82...	ff02::1:3	LLMNR	88	Standard query 0xa157 AAAA 05_17_08
11	0.204849	10.20.63.211	224.0.0.251	MDNS	166	Standard query 0x0000 PTR_companion-link_tcp.local, "QM" question PTR_rdlink_tcp.local, "QM" question PTR_NENCY's MacBook Air...
12	0.205062	fe80::1b83:39b:fdfe...	ff02::fb	MDNS	186	Standard query 0x0000 PTR_companion-link_tcp.local, "QM" question PTR_rdlink_tcp.local, "QM" question PTR_NENCY's MacBook Air...
13	0.205385	10.20.62.58	224.0.0.251	MDNS	102	Standard query 0x6d0e PTR_TWHH's Laptop_companion-link_tcp.local, "QM" question
17	0.307088	fe80::d405:5f99:3c6...	ff02::1:3	LLMNR	85	Standard query 0xfab6 A 10_04
18	0.307888	10.20.19.16	224.0.0.252	LLMNR	65	Standard query 0xfab6 A 10_04
19	0.307888	fe80::d405:5f99:3c6...	ff02::1:3	LLMNR	85	Standard query 0xfab6 AAAA 10_04
20	0.308197	10.20.19.16	224.0.0.252	LLMNR	65	Standard query 0xfab6 AAAA 10_04
21	0.308249	10.20.63.183	224.0.0.251	MDNS	117	Standard query response 0x0000 NSEC, cache flush TWHH's Laptop_companion-link_tcp.local
22	0.308607	fe80::845:3c59:67cc...	ff02::fb	MDNS	137	Standard query response 0x0000 NSEC, cache flush TWHH's Laptop_companion-link_tcp.local
30	0.409759	fe80::eeffa1e86:d28...	ff02::1:3	LLMNR	85	Standard query 0x1d7b A 10_04
31	0.409918	10.20.19.7	224.0.0.252	LLMNR	65	Standard query 0x1d7b A 10_04
32	0.410181	fe80::eeffa1e86:d28...	ff02::1:3	LLMNR	85	Standard query 0x5689 AAAA 10_04

Frame 11: 85 bytes on wire (680 bits), 85 bytes captured (680 bits) on interface DeviceNPF_{7E118522-08...}

Ethernet II, Src: HP_a4:44:2f (48:9e:bd:a4:44:2f), Dst: IPv6mcast_01:00:03 (33:33:00:01:00:03)

Internet Protocol Version 6, Src: fe80::19dc:7cce:8987:3f2b, Dst: ff02::1:3

User Datagram Protocol, Src Port: 54335, Dst Port: 5355

Link-local Multicast Name Resolution (query)

Packet 11 details:

```

  0000  33 33 00 01 00 03 48 9e  bd a4 44 2f 86 dd 60 04  33 33 00 01 00 03 48 9e
  0010  8e f0 00 1f 11 01 fe 80  00 00 00 00 00 00 19 dc  8e f0 00 1f 11 01 fe 80
  0020  7c ce 89 87 3f 2b ff 02  00 00 00 00 00 00 00 00  7c ce 89 87 3f 2b ff 02
  0030  00 00 01 00 03 d4 3f 14  eb 00 1f b2 84 9f 56 00  00 00 01 00 03 d4 3f 14
  0040  00 00 01 00 00 00 00 00  00 00 05 31 30 5f 30 34  00 00 01 00 00 00 00 00
  0050  00 00 01 00 01
  
```